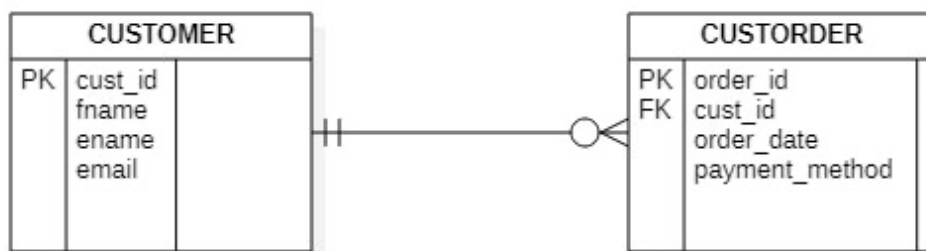


GIK2NV DSTM Lab 2: Information model for spatial data

During this lab, you will create an information model based on the concepts from the lecture: **Lec 04 DSMT (Spatial data)**. It is therefore very important that you have watched the lecture before you start the laboratory.

When creating the information model, I recommend that you use StarUML. This is a free program that contains many different features. **There is a short tutorial how to use StarUML on the course pages above.**

Use the Crow-foot notation for the model. Here is an example how it looks:



NOTE! This lab is also a group work lab.

To be approved in the lab, the following is required:

1. Hand in a well structured information model as a picture file.
2. You should record a movie with sound where you explain the model.

The task

Create an information model with the concept definitions from the lecture 04 (spatial data). You must include the following terms:

1. Route
2. Turn
3. Link
4. Node
5. Line (the geometry or topology of the link)
6. "Things" that are of interest to the users of the traffic network. for example liquer stores, banks or schools. You have to "connect" them to your model in a appropriate way.

And you have to use geographical coordinates in the model for it to work.

Things to think about before you start

DO NOT FORGET! "PK forms FK in the direction of the fork".

How do we store a coordinate? Metadata about which system?

If we paste the following into google: 60.48358252407589, 15.421925712733989

Then we end up in the dome roundabout with Google maps coordinate system WGS84.

The simplest is to store the coordinates in a column e.g.

WGS84_coord with values of the form: 60.48358252407589, 15.421925712733989.

Then we can put the coordinates as an attribute in tables where they are needed.

Example:

```
NODE-TABLE
node_id
WGS84_coord
```

A Link is defined as: "A link is the bird's eye view between two nodes" ie a vector or straight line.

The straight line equation from early high school math: $f(x) = kx + m$

If we go further and look at the TURN table.

A TURN object is defined using a PK (turn_id) and three columns that describe: entry_link, node and exit_link. Nothing else in the TURN table.

A ROUTE means that we, for example, travel from an address (coordinate) in Borlänge to an address (coordinate) Stockholm by car:

ROUTE

route_id = PK

maybe_a_name_on_the_route?

WGS84_start = Describes the start of the route

WGS84_end = Describes end

Question: What is the relationship between TURN and ROUTE?

This has important consequences for the model.

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